

China: Data centers hobbled by chips and planning gaps

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China's data center buildout hobbled by chip shortages and planning missteps

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Uncoordinated regional investment under Beijing's East Data, West Compute (EDWC) strategy created data centers without adequate fiber connectivity or local talent pools, leaving many facilities underused.

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US export controls have cut off China's access to advanced chips for frontier AI training, capping high-end compute capacity and forcing some Chinese AI labs to shift model training overseas through cloud services.

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China is taking a cautious approach to data center investment, but a meaningful domestic chip breakthrough would quickly shift the calculus by lifting compute constraints, increasing demand, and strengthening the case for a broader buildout.

Uncoordinated planning has resulted in a glut of underused data center facilities. Starting in the 2010s, western provinces courted data center investment, an effort Beijing formalized in 2022 as the EDWC initiative. Designed to shift data center investment from the congested eastern seaboard to interior hubs, EDWC leverages the west's cheap land, abundant energy, and cooler climates to catalyze economic growth. It also aims to match compute supply and demand between inland and coastal China while mitigating environmental and social costs—some of the same challenges the US faces in its own data center buildout.

Motivated by the economic growth potential of infrastructure projects, provincial and local governments responded enthusiastically to the initiative and competed to build data centers with little additional coordination. Ningxia, Guizhou, and Xinjiang rushed to build data center parks with the same speculative real-estate logic: new infrastructure without matching demand. The state-backed Science and Technology Daily reported in July 2025 that one western compute hub had rack utilization below 50% and actual server utilization below 30%—a figure common among data centers near tier-three and tier-four cities.

Western China data centers are not suited for AI-era workloads. The geographic shift introduces a regulatory constraint: Unlike US hyperscalers, Chinese cloud companies cannot build or own fiber networks outside their own data centers: The national telecom regulator awards operating licenses only to government-backed telecoms. All data traffic from the eastern coast must transit through three large

state-owned telecom companies, adding layers of permissions and latency. Delayed response has thus been a major barrier to EDWC's deployment. Talent is another impediment: According to one industry report, 87.5% of China's data center talent, including electricians, cooling specialists, and network engineers, remains concentrated in northern, eastern, and southern parts of the country. Some companies fly their engineers across the country monthly to service western facilities at high cost.

The combined effect of deficient coordination and resource mismatches limits western China's centers largely to "cold data" storage—archival content that tolerates slow retrieval. For "hot data" demanding low latency and high throughput, distant data centers are an increasingly impractical solution. In the AI age, inference requires low latency, making the western China data center model structurally ill-suited to serving eastern users in real time. Inner Mongolia is a relative success story. Hohhot and Ulanqab's proximity to Beijing partially addresses the latency problems that plagued facilities in Guizhou and Ningxia, making it the best option for data center investment.

Moreover, the data centers lack the chips needed for frontier AI training. US export restrictions in place since October 2022 have cut off China's access to the advanced semiconductors needed to train frontier models. Chinese chipmakers have been slow to catch up with industry leaders. Although domestic suppliers such as Huawei promote their chips as competitive with Nvidia's, China's most advanced models still heavily rely on Nvidia chips for training. Meanwhile, Huawei's chips mainly serve inference, though the chipmaker struggles with low manufacturing yields. The resulting chip shortages have held back China's high-end compute capacity and pushed Chinese AI labs to move some training overseas, via cloud services.

Chip controls have shaped a cautious approach to new data center investment. China has installed less than two-thirds of US compute power. The slower growth pace reflects the structural ceiling imposed by chip controls and the lessons from the previous infrastructure cycle. In June 2026, Beijing announced a \$295 billion five-year program for data center construction—primarily through state telecom operators using domestic AI chips. The new program likely seeks to build on Inner Mongolia's success and adopt China's advanced chips to meet the growing demand for AI inference. It also seeks to use government support to spur progress among Chinese chipmakers. Yet the investment commitment pales in comparison to the scale of US plans: The top five American cloud companies are on track to spend nearly \$700 billion in 2026 alone.

Progress on chip capacity can rapidly shift China's data center strategy. Absent a chip breakthrough, scaling data centers to US levels would only worsen the overcapacity Beijing is already trying to manage, leaving little incentive for the government to address its planning problems. However, if Chinese chipmakers achieve meaningful advances, the calculus would shift quickly, as data center demand would rapidly increase.

Xiaomeng Lu

Director, Geo-technology

+1 202.316.7259

lu@eurasiagroup.net

Heather West

Practice Head, Geo-technology

west@eurasiagroup.net

EG Geo-technology
geo-technology@eurasiagroup.net

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